

CATEGORIES

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1. DEFINITIONS

We recall the definitions, partly to fix notation.

Definition 1.1. A *category* $\mathcal{C}$ consists of the following data:

- (1) A set of objects $\text{Ob}(\mathcal{C})$.
- (2) For each pair $x, y \in \text{Ob}(\mathcal{C})$ a set of morphisms $\text{Mor}_{\mathcal{C}}(x, y)$.
- (3) For each triple $x, y, z \in \text{Ob}(\mathcal{C})$ a composition map $\text{Mor}_{\mathcal{C}}(y, z) \times \text{Mor}_{\mathcal{C}}(x, y) \rightarrow \text{Mor}_{\mathcal{C}}(x, z)$, denoted $(\phi, \psi) \mapsto \phi \circ \psi$.

These data are to satisfy the following rules:

- (1) For every element $x \in \text{Ob}(\mathcal{C})$ there exists a morphism $\text{id}_x \in \text{Mor}_{\mathcal{C}}(x, x)$ such that $\text{id}_x \circ \phi = \phi$ and $\psi \circ \text{id}_x = \psi$ whenever these compositions make sense.
- (2) Composition is associative, i.e., $(\phi \circ \psi) \circ \chi = \phi \circ (\psi \circ \chi)$ whenever these compositions make sense.

Definition 1.2. A *functor* $F : \mathcal{A} \rightarrow \mathcal{B}$ between two categories $\mathcal{A}, \mathcal{B}$ is given by the following data:

- (1) A map $F : \text{Ob}(\mathcal{A}) \rightarrow \text{Ob}(\mathcal{B})$.
- (2) For every $x, y \in \text{Ob}(\mathcal{A})$ a map $F : \text{Mor}_{\mathcal{A}}(x, y) \rightarrow \text{Mor}_{\mathcal{B}}(F(x), F(y))$, denoted $\phi \mapsto F(\phi)$.

These data should be compatible with composition and identity morphisms in the following manner: $F(\phi \circ \psi) = F(\phi) \circ F(\psi)$ for a composable pair (ϕ, ψ) of morphisms of $\mathcal{A}$ and $F(\text{id}_x) = \text{id}_{F(x)}$.

Note that every category $\mathcal{A}$ has an *identity* functor $\text{id}_{\mathcal{A}}$. In addition, given a functor $G : \mathcal{B} \rightarrow \mathcal{C}$ and a functor $F : \mathcal{A} \rightarrow \mathcal{B}$ there is a *composition* functor $G \circ F : \mathcal{A} \rightarrow \mathcal{C}$ defined in an obvious manner.

Definition 1.3. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a functor.

- (1) We say F is *faithful* if for any objects $x, y \in \text{Ob}(\mathcal{A})$ the map

$$F : \text{Mor}_{\mathcal{A}}(x, y) \rightarrow \text{Mor}_{\mathcal{B}}(F(x), F(y))$$

is injective.

- (2) If these maps are all bijective then F is called *fully faithful*.

- (3) The functor F is called *essentially surjective* if for any object $y \in \text{Ob}(\mathcal{B})$ there exists an object $x \in \text{Ob}(\mathcal{A})$ such that $F(x)$ is isomorphic to y in $\mathcal{B}$.

Definition 1.4. A *subcategory* of a category $\mathcal{B}$ is a category $\mathcal{A}$ whose objects and arrows form subsets of the objects and arrows of $\mathcal{B}$ and such that source, target and composition in $\mathcal{A}$ agree with those of $\mathcal{B}$ and such that the identity morphism of an object of $\mathcal{A}$ matches the one in $\mathcal{B}$. We say $\mathcal{A}$ is a *full subcategory* of $\mathcal{B}$ if $\text{Mor}_{\mathcal{A}}(x, y) = \text{Mor}_{\mathcal{B}}(x, y)$ for all $x, y \in \text{Ob}(\mathcal{A})$. We say $\mathcal{A}$ is a *strictly full subcategory* of $\mathcal{B}$ if it is a full subcategory and given $x \in \text{Ob}(\mathcal{A})$ any object of $\mathcal{B}$ which is isomorphic to x is also in $\mathcal{A}$.

If $\mathcal{A} \subset \mathcal{B}$ is a subcategory then the identity map is a functor from $\mathcal{A}$ to $\mathcal{B}$. Furthermore a subcategory $\mathcal{A} \subset \mathcal{B}$ is full if and only if the inclusion functor is fully faithful. Note that given a category $\mathcal{B}$ the set of full subcategories of $\mathcal{B}$ is the same as the set of subsets of $\text{Ob}(\mathcal{B})$.

Example 1.5. Given a category $\mathcal{C}$ and an object $X \in \text{Ob}(\mathcal{C})$ we define the *category of objects over X* , denoted $\mathcal{C}/X$ as follows. The objects of $\mathcal{C}/X$ are morphisms $Y \rightarrow X$ for some $Y \in \text{Ob}(\mathcal{C})$. Morphisms between objects $Y \rightarrow X$ and $Y' \rightarrow X$ are morphisms $Y \rightarrow Y'$ in $\mathcal{C}$ that make the obvious diagram commute. Note that there is a functor $p_X : \mathcal{C}/X \rightarrow \mathcal{C}$ which simply forgets the morphism. Moreover given a morphism $f : X' \rightarrow X$ in $\mathcal{C}$ there is an induced functor $F : \mathcal{C}/X' \rightarrow \mathcal{C}/X$ obtained by composition with f , and $p_X \circ F = p_{X'}$.

Example 1.6. Given a category $\mathcal{C}$ and an object $X \in \text{Ob}(\mathcal{C})$ we define the *category of objects under X* , denoted $X/\mathcal{C}$ as follows. The objects of $X/\mathcal{C}$ are morphisms $X \rightarrow Y$ for some $Y \in \text{Ob}(\mathcal{C})$. Morphisms between objects $X \rightarrow Y$ and $X \rightarrow Y'$ are morphisms $Y \rightarrow Y'$ in $\mathcal{C}$ that make the obvious diagram commute. Note that there is a functor $p_X : X/\mathcal{C} \rightarrow \mathcal{C}$ which simply forgets the morphism. Moreover given a morphism $f : X' \rightarrow X$ in $\mathcal{C}$ there is an induced functor $F : X/\mathcal{C} \rightarrow X'/\mathcal{C}$ obtained by composition with f , and $p_{X'} \circ F = p_X$.

Definition 1.7. Two categories are *equivalent* if there exist functors from one to the other and backwards which composed with each other are the corresponding identities.

Lemma 1.8. A functor is an equivalence of categories if and only if it is both fully faithful and essentially surjective.

Proof. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be essentially surjective and fully faithful. As by convention all categories are small and as F is essentially surjective we can, using the axiom of choice, choose for every $X \in \text{Ob}(\mathcal{B})$ an object $j(X)$ of $\mathcal{A}$ and an isomorphism $i_X : X \rightarrow F(j(X))$. Then we apply Lemma ?? using that F is fully faithful. $\square$

2. MONOMORPHISMS

Definition 2.1. Let $\mathcal{C}$ be a category and let $f : X \rightarrow Y$ be a morphism of $\mathcal{C}$.

- (1) We say that f is a *monomorphism* if for every object W and every pair of morphisms $a, b : W \rightarrow X$ such that $f \circ a = f \circ b$ we have $a = b$.
- (2) We say that f is an *epimorphism* if for every object W and every pair of morphisms $a, b : Y \rightarrow W$ such that $a \circ f = b \circ f$ we have $a = b$.

Definition 2.2. Let $\mathcal{C}$ be a category, and let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a map of presheaves of sets.

- (1) We say that φ is *injective* if for every object U of $\mathcal{C}$ the map $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective.
- (2) We say that φ is *surjective* if for every object U of $\mathcal{C}$ the map $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is surjective.

Lemma 2.3. *The injective (resp. surjective) maps defined above are exactly the monomorphisms (resp. epimorphisms) of $PSh(\mathcal{C})$. A map is an isomorphism if and only if it is both injective and surjective.*

3. PRESHEAVES

Definition 3.1. A *presheaf of sets* on $\mathcal{C}$ is a contravariant functor from $\mathcal{C}$ to *Sets*. *Morphisms of presheaves* are natural transformations of functors. The category of presheaves of sets is denoted $PSh(\mathcal{C})$ or $\hat{\mathcal{C}}$.

4. YONEDA LEMMA

The Yoneda lemma says that the sections over a of a presheaf X can be described completely as natural transformations between the presheaf $\text{Hom}(-, a)$ and X .

Definition 4.1. Let A be a category. The *Yoneda embedding* is the functor

$$h : A \rightarrow \hat{A}$$

whose value at an object a of A is the presheaf

$$h_a = \text{Hom}_A(-, a).$$

In other words, the evaluation of the presheaf h_a at an object c of A is the set of maps from c to a .

Theorem 4.2 (Yoneda lemma). *For any presheaf X over A , there is a natural bijection (in *Sets* I think!!)*

$$\begin{aligned} \text{Hom}_{\hat{A}}(h_a, X) &\xrightarrow{\cong} X_a \\ (h_a \xrightarrow{u} X) &\longmapsto u_a(1_a) \end{aligned}$$

5. INTERNAL HOM

Upshot. Internal Hom is when the Hom set of two objects in some category is in also an object of the category. Down-to-earth, that for two sheaves $\mathcal{F}, \mathcal{G}$, $U \mapsto \text{Hom}(\mathcal{F}(U), \mathcal{G}(U))$ is also a sheaf, called $\mathcal{H}om$.

I start with Stacks Project approach.

Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$ -modules. Consider the rule

$$U \longmapsto \text{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U).$$

It follows from the discussion in Sheaves, Section ?? that this is a sheaf of abelian groups. In addition, given an element $\varphi \in \text{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U)$ and a section $f \in \mathcal{O}_X(U)$ then we can define $f\varphi \in \text{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U)$ by either precomposing with multiplication by f on $\mathcal{F}|_U$ or postcomposing with multiplication by f on $\mathcal{G}|_U$ (it gives the same result). Hence we in fact get a sheaf of $\mathcal{O}_X$ -modules. We will denote this sheaf $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$. There is a canonical “evaluation” morphism

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathcal{G}.$$

For every $x \in X$ there is also a canonical morphism

$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})_x \rightarrow \text{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$$

which is rarely an isomorphism.

Cartesian closed category In the category of sets there is a bijection $\text{Hom}(X \times Y, Z) \cong \text{Hom}(X, \text{Hom}(Y, Z))$ that depends naturally on X , Y and Z . The notions related to this bijection are “Cartesian closed category”, “currying” and “internal Hom”.

Definition 5.1. A category $\mathcal{C}$ is *Cartesian closed* if:

- (1) $\mathcal{C}$ has all finite products (Caveat: some require that $\mathcal{C}$ has all finite limits)
- (2) For any object Y the functor $- \times Y$ has a right adjoint, which we will denote by $\text{Map}(Y, -)$ or by $-^Y$.

Remark 5.2. By section 3 here, the second property above implies that we get a functor $\text{Map}(-, -) : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathcal{C}$, and moreover we get natural isomorphisms $\text{Hom}(X, \text{Map}(Y, Z)) \cong \text{Hom}(X \times Y, Z)$ and $\text{Map}(X, \text{Map}(Y, Z)) \cong \text{Map}(X \times Y, Z)$.

6. PRODUCT

Definition 6.1. A *product* in a category $\mathcal{C}$ is

$$\begin{array}{ccc} P & \xrightarrow{a} & A \\ b \downarrow & & \\ B & & \end{array}$$

such that for every other

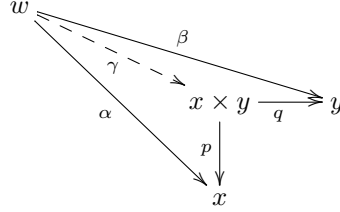
$$\begin{array}{ccc} P' & \xrightarrow{a'} & A \\ b' \downarrow & & \\ B & & \end{array}$$

there exists a unique map $p : P' \rightarrow P$ such that $ap = a'$ and $bp = b'$.

For completeness here’s Stacks Project formulation:

Definition 6.2. Let $x, y \in \text{Ob}(\mathcal{C})$. A *product* of x and y is an object $x \times y \in \text{Ob}(\mathcal{C})$ together with morphisms $p \in \text{Mor}_{\mathcal{C}}(x \times y, x)$ and $q \in \text{Mor}_{\mathcal{C}}(x \times y, y)$ such that the following universal property holds: for any $w \in \text{Ob}(\mathcal{C})$ and morphisms

$\alpha \in \text{Mor}_{\mathcal{C}}(w, x)$ and $\beta \in \text{Mor}_{\mathcal{C}}(w, y)$ there is a unique $\gamma \in \text{Mor}_{\mathcal{C}}(w, x \times y)$ making the diagram



commute.

And some nice piece of information and a definition also from Stacks Project.

If a product exists it is unique up to unique isomorphism. This follows from the Yoneda lemma as the definition requires $x \times y$ to be an object of $\mathcal{C}$ such that

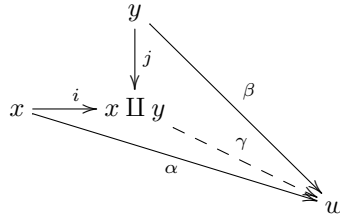
$$h_{x \times y}(w) = h_x(w) \times h_y(w)$$

functorially in w . In other words the product $x \times y$ is an object representing the functor $w \mapsto h_x(w) \times h_y(w)$.

Definition 6.3. We say the category $\mathcal{C}$ has *products of pairs of objects* if a product $x \times y$ exists for any $x, y \in \text{Ob}(\mathcal{C})$.

7. COPRODUCTS OF PAIRS

Definition 7.1. Let $x, y \in \text{Ob}(\mathcal{C})$. A *coproduct*, or *amalgamated sum* of x and y is an object $x \amalg y \in \text{Ob}(\mathcal{C})$ together with morphisms $i \in \text{Mor}_{\mathcal{C}}(x, x \amalg y)$ and $j \in \text{Mor}_{\mathcal{C}}(y, x \amalg y)$ such that the following universal property holds: for any $w \in \text{Ob}(\mathcal{C})$ and morphisms $\alpha \in \text{Mor}_{\mathcal{C}}(x, w)$ and $\beta \in \text{Mor}_{\mathcal{C}}(y, w)$ there is a unique $\gamma \in \text{Mor}_{\mathcal{C}}(x \amalg y, w)$ making the diagram



commute.

If a coproduct exists it is unique up to unique isomorphism. This follows from the Yoneda lemma (applied to the opposite category) as the definition requires $x \amalg y$ to be an object of $\mathcal{C}$ such that

$$\text{Mor}_{\mathcal{C}}(x \amalg y, w) = \text{Mor}_{\mathcal{C}}(x, w) \times \text{Mor}_{\mathcal{C}}(y, w)$$

functorially in w .

Definition 7.2. We say the category $\mathcal{C}$ has *coproducts of pairs of objects* if a coproduct $x \amalg y$ exists for any $x, y \in \text{Ob}(\mathcal{C})$.

8. FIBRE PRODUCTS

Definition 8.1. Let $x, y, z \in \text{Ob}(\mathcal{C})$, $f \in \text{Mor}_{\mathcal{C}}(x, y)$ and $g \in \text{Mor}_{\mathcal{C}}(z, y)$. A *fibre product* of f and g is an object $x \times_y z \in \text{Ob}(\mathcal{C})$ together with morphisms $p \in \text{Mor}_{\mathcal{C}}(x \times_y z, x)$ and $q \in \text{Mor}_{\mathcal{C}}(x \times_y z, z)$ making the diagram

$$\begin{array}{ccc} x \times_y z & \xrightarrow{q} & z \\ p \downarrow & & \downarrow g \\ x & \xrightarrow{f} & y \end{array}$$

commute, and such that the following universal property holds: for any $w \in \text{Ob}(\mathcal{C})$ and morphisms $\alpha \in \text{Mor}_{\mathcal{C}}(w, x)$ and $\beta \in \text{Mor}_{\mathcal{C}}(w, z)$ with $f \circ \alpha = g \circ \beta$ there is a unique $\gamma \in \text{Mor}_{\mathcal{C}}(w, x \times_y z)$ making the diagram

$$\begin{array}{ccccc} w & & & & \\ & \searrow \beta & & \searrow & \\ & & x \times_y z & \xrightarrow{q} & z \\ & \swarrow \alpha & \downarrow p & & \downarrow g \\ & & x & \xrightarrow{f} & y \end{array}$$

commute.

Remark 8.2. I think that a product is a fibre product when we put y to be the terminal object!

If a fibre product exists it is unique up to unique isomorphism. This follows from the Yoneda lemma as the definition requires $x \times_y z$ to be an object of $\mathcal{C}$ such that

$$h_{x \times_y z}(w) = h_x(w) \times_{h_y(w)} h_z(w)$$

functorially in w . In other words the fibre product $x \times_y z$ is an object representing the functor $w \mapsto h_x(w) \times_{h_y(w)} h_z(w)$.

Definition 8.3. We say a commutative diagram

$$\begin{array}{ccc} w & \longrightarrow & z \\ \downarrow & & \downarrow \\ x & \longrightarrow & y \end{array}$$

in a category is *cartesian* if w and the morphisms $w \rightarrow x$ and $w \rightarrow z$ form a fibre product of the morphisms $x \rightarrow y$ and $z \rightarrow y$.

Definition 8.4. We say the category $\mathcal{C}$ *has fibre products* if the fibre product exists for any $f \in \text{Mor}_{\mathcal{C}}(x, y)$ and $g \in \text{Mor}_{\mathcal{C}}(z, y)$.

Definition 8.5. A morphism $f : x \rightarrow y$ of a category $\mathcal{C}$ is said to be *representable* if for every morphism $z \rightarrow y$ in $\mathcal{C}$ the fibre product $x \times_y z$ exists.

Lemma 8.6. Let $\mathcal{C}$ be a category. Let $f : x \rightarrow y$, and $g : y \rightarrow z$ be representable. Then $g \circ f : x \rightarrow z$ is representable.

Proof. Let $t \in \text{Ob}(\mathcal{C})$ and $\varphi \in \text{Mor}_{\mathcal{C}}(t, z)$. As g and f are representable, we obtain commutative diagrams

$$\begin{array}{ccc} y \times_z t & \xrightarrow{q} & t \\ p \downarrow & & \downarrow \varphi \\ y & \xrightarrow{g} & z \end{array} \quad \begin{array}{ccc} x \times_y (y \times_z t) & \xrightarrow{q'} & y \times_z t \\ p' \downarrow & & \downarrow p \\ x & \xrightarrow{f} & y \end{array}$$

with the universal property of Definition 8.1. We claim that $x \times_z t = x \times_y (y \times_z t)$ with morphisms $q \circ q' : x \times_z t \rightarrow t$ and $p' : x \times_z t \rightarrow x$ is a fibre product. First, it follows from the commutativity of the diagrams above that $\varphi \circ q \circ q' = g \circ f \circ p'$. To verify the universal property, let $w \in \text{Ob}(\mathcal{C})$ and suppose $\alpha : w \rightarrow x$ and $\beta : w \rightarrow y$ are morphisms with $\varphi \circ \beta = g \circ f \circ \alpha$. By definition of the fibre product, there are unique morphisms δ and γ such that

$$\begin{array}{ccccc} w & & & & \\ & \searrow \beta & & \searrow \delta & \\ & & y \times_z t & \xrightarrow{q} & t \\ & \searrow f \circ \alpha & \downarrow p & & \downarrow \varphi \\ & & y & \xrightarrow{g} & z \end{array}$$

and

$$\begin{array}{ccccc} w & & & & \\ & \searrow \delta & & \searrow \gamma & \\ & & x \times_y (y \times_z t) & \xrightarrow{q'} & y \times_z t \\ & \searrow \alpha & \downarrow p' & & \downarrow p \\ & & x & \xrightarrow{f} & y \end{array}$$

commute. Then, γ makes the diagram

$$\begin{array}{ccccc} w & & & & \\ & \searrow \beta & & \searrow \gamma & \\ & & x \times_z t & \xrightarrow{q \circ q'} & t \\ & \searrow \alpha & \downarrow p' & & \downarrow \varphi \\ & & x & \xrightarrow{g \circ f} & z \end{array}$$

commute. To show its uniqueness, let γ' verify $q \circ q' \circ \gamma' = \beta$ and $p' \circ \gamma' = \alpha$. Because γ is unique, we just need to prove that $q' \circ \gamma' = \delta$ and $p' \circ \gamma' = \alpha$ to conclude. We supposed the second equality. For the first one, we also need to use the uniqueness of delta. Notice that δ is the only morphism verifying $q \circ \delta = \beta$ and $p \circ \delta = f \circ \alpha$. We already supposed that $q \circ (q' \circ \gamma') = \beta$. Furthermore, by definition of the fibre product, we know that $f \circ p' = p \circ q'$. Therefore:

$$p \circ (q' \circ \gamma') = (p \circ q') \circ \gamma' = (f \circ p') \circ \gamma' = f \circ (p' \circ \gamma') = f \circ \alpha.$$

Then $q' \circ \gamma' = \delta$, which concludes the proof. $\square$

Lemma 8.7. *Let $\mathcal{C}$ be a category. Let $f : x \rightarrow y$ be representable. Let $y' \rightarrow y$ be a morphism of $\mathcal{C}$. Then the morphism $x' := x \times_y y' \rightarrow y'$ is representable also.*

Proof. Let $z \rightarrow y'$ be a morphism. The fibre product $x' \times_{y'} z$ is supposed to represent the functor

$$\begin{aligned} w &\mapsto h_{x'}(w) \times_{h_{y'}(w)} h_z(w) \\ &= (h_x(w) \times_{h_y(w)} h_{y'}(w)) \times_{h_{y'}(w)} h_z(w) \\ &= h_x(w) \times_{h_y(w)} h_z(w) \end{aligned}$$

which is representable by assumption. $\square$

9. EXAMPLES OF FIBRE PRODUCTS

In this section we list examples of fibre products and we describe them.

As a really trivial first example we observe that the category of sets has fibre products and hence every morphism is representable. Namely, if $f : X \rightarrow Y$ and $g : Z \rightarrow Y$ are maps of sets then we define $X \times_Y Z$ as the subset of $X \times Z$ consisting of pairs (x, z) such that $f(x) = g(z)$. The morphisms $p : X \times_Y Z \rightarrow X$ and $q : X \times_Y Z \rightarrow Z$ are the projection maps $(x, z) \mapsto x$, and $(x, z) \mapsto z$. Finally, if $\alpha : W \rightarrow X$ and $\beta : W \rightarrow Z$ are morphisms such that $f \circ \alpha = g \circ \beta$ then the map $W \rightarrow X \times_Y Z$, $w \mapsto (\alpha(w), \beta(w))$ obviously ends up in $X \times_Y Z$ as desired.

In many categories whose objects are sets endowed with certain types of algebraic structures the fibre product of the underlying sets also provides the fibre product in the category. For example, suppose that X, Y and Z above are groups and that f, g are homomorphisms of groups. Then the set-theoretic fibre product $X \times_Y Z$ inherits the structure of a group, simply by defining the product of two pairs by the formula $(x, z) \cdot (x', z') = (xx', zz')$. Here we list those categories for which a similar reasoning works.

- (1) The category *Groups* of groups.
- (2) The category *G-Sets* of sets endowed with a left G -action for some fixed group G .
- (3) The category of rings.
- (4) The category of R -modules given a ring R .

10. PUSHOUTS

The dual notion to fibre products is that of pushouts.

Definition 10.1. Let $x, y, z \in \text{Ob}(\mathcal{C})$, $f \in \text{Mor}_{\mathcal{C}}(y, x)$ and $g \in \text{Mor}_{\mathcal{C}}(y, z)$. A *pushout* of f and g is an object $x \amalg_y z \in \text{Ob}(\mathcal{C})$ together with morphisms $p \in \text{Mor}_{\mathcal{C}}(x, x \amalg_y z)$ and $q \in \text{Mor}_{\mathcal{C}}(z, x \amalg_y z)$ making the diagram

$$\begin{array}{ccc} y & \xrightarrow{g} & z \\ f \downarrow & & \downarrow q \\ x & \xrightarrow{p} & x \amalg_y z \end{array}$$

commute, and such that the following universal property holds: For any $w \in \text{Ob}(\mathcal{C})$ and morphisms $\alpha \in \text{Mor}_{\mathcal{C}}(x, w)$ and $\beta \in \text{Mor}_{\mathcal{C}}(z, w)$ with $\alpha \circ f = \beta \circ g$ there is a

unique $\gamma \in \text{Mor}_C(x \amalg_y z, w)$ making the diagram

$$\begin{array}{ccc}
 y & \xrightarrow{g} & z \\
 f \downarrow & & \downarrow q \\
 x & \xrightarrow{p} & x \amalg_y z \\
 & \searrow \alpha & \searrow \gamma \\
 & & w
 \end{array}
 \quad \beta$$

commute.

It is possible and straightforward to prove the uniqueness of the triple $(x \amalg_y z, p, q)$ up to unique isomorphism (if it exists) by direct arguments. Another possibility is to think of the pushout as the fibre product in the opposite category, thereby getting this uniqueness for free from the discussion in Section 8.

Definition 10.2. We say a commutative diagram

$$\begin{array}{ccc}
 y & \longrightarrow & z \\
 \downarrow & & \downarrow \\
 x & \longrightarrow & w
 \end{array}$$

in a category is *cocartesian* if w and the morphisms $x \rightarrow w$ and $z \rightarrow w$ form a pushout of the morphisms $y \rightarrow x$ and $y \rightarrow z$.

According to Sergei Burkin, another way to define a pushout is as follows. The diagram

$$\begin{array}{ccc}
 A & \longrightarrow & B \\
 \downarrow & \lrcorner & \downarrow \\
 C & \longrightarrow & D
 \end{array}$$

is a *pushout* if the diagram

$$\begin{array}{ccc}
 \text{Map}(D, X) & \longrightarrow & \text{Map}(B, X) \\
 \downarrow & \lrcorner & \downarrow \\
 \text{Map}(C, D) & \longrightarrow & \text{Map}(D, X)
 \end{array}$$

is a pullback.

In fact, spheres are (homotopy?) pushouts:

$$\begin{array}{ccc}
 S^{n-1} & \longrightarrow & \text{pt} \\
 \downarrow & \lrcorner & \downarrow \\
 \text{pt} & \longrightarrow & S^n
 \end{array}$$

which is homotopically the same as

$$\begin{array}{ccc}
 S^{n-1} & \longrightarrow & D^n \\
 \downarrow & \lrcorner & \downarrow \\
 D^n & \longrightarrow & S^n,
 \end{array}$$

the familiar construction of glueing two disks along their boundary to obtain a sphere.

11. LIMITS AND COLIMITS

A colimit is: take an abstract diagram $\mathcal{I}$. Realize it in the category that interests you (for the stack this are open sets which contain the point, and are a poset by inclusion). The colimit is the unique object that there is an arrow from all the points of the diagram and satisfies some universal property that any other such object there's a morphism bla bla. (In the case of the stack it is the unique open set contained in all the other open sets. Which actually it's interesting that such a thing exists.)

A product is actually a particular case of a limit.

Let $\mathcal{C}$ be a category. A *diagram* in $\mathcal{C}$ is simply a functor $M : \mathcal{I} \rightarrow \mathcal{C}$. We say that $\mathcal{I}$ is the *index category* or that M is an $\mathcal{I}$ -diagram. We will use the notation M_i to denote the image of the object i of $\mathcal{I}$. Hence for $\phi : i \rightarrow i'$ a morphism in $\mathcal{I}$ we have $M(\phi) : M_i \rightarrow M_{i'}$.

Definition 11.1. A *limit* of the $\mathcal{I}$ -diagram M in the category $\mathcal{C}$ is given by an object $\lim_{\mathcal{I}} M$ in $\mathcal{C}$ together with morphisms $p_i : \lim_{\mathcal{I}} M \rightarrow M_i$ such that

- (1) for $\phi : i \rightarrow i'$ a morphism in $\mathcal{I}$ we have $p_{i'} = M(\phi) \circ p_i$, and
- (2) for any object W in $\mathcal{C}$ and any family of morphisms $q_i : W \rightarrow M_i$ (indexed by $i \in \text{Ob}(\mathcal{I})$) such that for all $\phi : i \rightarrow i'$ in $\mathcal{I}$ we have $q_{i'} = M(\phi) \circ q_i$ there exists a unique morphism $q : W \rightarrow \lim_{\mathcal{I}} M$ such that $q_i = p_i \circ q$ for every object i of $\mathcal{I}$.

Limits $(\lim_{\mathcal{I}} M, (p_i)_{i \in \text{Ob}(\mathcal{I})})$ are (if they exist) unique up to unique isomorphism by the uniqueness requirement in the definition. Products of pairs, fibre products, and equalizers are examples of limits. The limit over the empty diagram is a final object of $\mathcal{C}$. In the category of sets all limits exist. The dual notion is that of colimits.

Definition 11.2. A *colimit* of the $\mathcal{I}$ -diagram M in the category $\mathcal{C}$ is given by an object $\text{colim}_{\mathcal{I}} M$ in $\mathcal{C}$ together with morphisms $s_i : M_i \rightarrow \text{colim}_{\mathcal{I}} M$ such that

- (1) for $\phi : i \rightarrow i'$ a morphism in $\mathcal{I}$ we have $s_i = s_{i'} \circ M(\phi)$, and
- (2) for any object W in $\mathcal{C}$ and any family of morphisms $t_i : M_i \rightarrow W$ (indexed by $i \in \text{Ob}(\mathcal{I})$) such that for all $\phi : i \rightarrow i'$ in $\mathcal{I}$ we have $t_i = t_{i'} \circ M(\phi)$ there exists a unique morphism $t : \text{colim}_{\mathcal{I}} M \rightarrow W$ such that $t_i = t \circ s_i$ for every object i of $\mathcal{I}$.

Definition 11.3. Suppose that I is a set, and suppose given for every $i \in I$ an object M_i of the category $\mathcal{C}$. A *product* $\prod_{i \in I} M_i$ is by definition $\lim_{\mathcal{I}} M$ (if it exists) where $\mathcal{I}$ is the category having only identities as morphisms and having the elements of I as objects.

An important special case is where $I = \emptyset$ in which case the product is a final object of the category. The morphisms $p_i : \prod M_i \rightarrow M_i$ are called the *projection morphisms*.

Definition 11.4. Suppose that I is a set, and suppose given for every $i \in I$ an object M_i of the category $\mathcal{C}$. A *coproduct* $\coprod_{i \in I} M_i$ is by definition $\text{colim}_{\mathcal{I}} M$ (if it exists) where $\mathcal{I}$ is the category having only identities as morphisms and having the elements of I as objects.

An important special case is where $I = \emptyset$ in which case the coproduct is an initial object of the category. Note that the coproduct comes equipped with morphisms $M_i \rightarrow \coprod M_i$. These are sometimes called the *coprojections*.

There is also so-called statement of “commutativity of limits”.

12. CATEGORIES FIBRED IN GROUPOIDS

General fibration: every map on the base has a lift that is p -cartesian. (Discrete fibration: unique?) Fibered in groupoids: all maps in the Étale space are p -cartesian.

And to my surprise, these are none other than categories fibered in groupoids according to Stacks Project 003S section on Categories fibered in groupoids.

Let $p : \mathcal{E} \rightarrow \mathcal{B}$ be any functor. The idea of fibrations is to lift maps from $\mathcal{B}$ to $\mathcal{E}$. We will fix the target; the condition is that for maps on $\mathcal{B}$ with target $p(y)$ there exist (unique? that’s the condition for discrete fibrations) maps on $\mathcal{E}$ corresponding via p . Why fix the target? Because it’s like a base-change. But not only that: we need that the map that exists is good; it has to be p -Cartesian:

Definition 12.1. Let $p : \mathcal{E} \rightarrow \mathcal{B}$. A morphism f in $\mathcal{E}$ is *p -cartesian* if for all g and for all h' such that $p(g) = p(f) \circ h'$ there exists a unique h such that $p(h) = h'$ and $g = f \circ h$

$$\begin{array}{ccc} Z & & \\ \downarrow h & \searrow g & \\ X & \xrightarrow{f} & Y \end{array}$$

given that

$$\begin{array}{ccc} p(Z) & & \\ \downarrow h' & \searrow p(g) & \\ p(X) & \xrightarrow{p(f)} & p(Y). \end{array}$$

Definition 12.2. $p : \mathcal{E} \rightarrow \mathcal{B}$ is a *fibration* if for all Y and all $f' : X' \rightarrow p(Y)$ there exists X and $f : X \rightarrow Y$ such that

- (1) $p(f) = f'$, and
- (2) f is p -Cartesian.

Recall Definition ?? of discrete fibration: $p : \mathcal{E} \rightarrow \mathcal{B}$ is a discrete fibration if for all $Y \in \mathcal{E}$ and for all $f' : X' \rightarrow p(Y)$ there exist **unique** $(X \text{ and } f : X \rightarrow Y)$ such that $p(f) = f'$.

Lemma 12.3. *If $p : \mathcal{E} \rightarrow \mathcal{B}$ is a discrete fibration then it is a fibration, i.e. f is also p -Cartesian.*

Proof. Uniqueness ensures source and target coincide in all lifts. □

Lemma 12.4. *Composition of p -Cartesian morphisms is p -Cartesian.*

Proof. Write the diagrams and don’t forget to prove uniqueness. □

Lemma 12.5. *2-for-3 for p -Cartesian.*

Proof. Write the diagrams and don’t forget to prove uniqueness. □

The notion of pseudofunctor $F : \mathcal{B}^{\text{op}} \rightarrow \mathbf{Cat}$ is like that of a functor but we ask that $F(\text{id}_A) \cong \text{id}_{F(A)}$ and $F(f) \circ F(g) \cong F(f \circ g)$ in some weaker sense than equality. (This is very much like it happens in ∞ -categories.) It turns out that

Fibrations over $\mathcal{B} \rightleftarrows$ pseudo functors $F : \mathcal{B}^{\text{op}} \rightarrow \mathbf{Cat}$.

Definition 12.6. A functor $p : \mathcal{E} \rightarrow \mathcal{B}$ is called a *fibration in groupoids* if it is a fibration and every fiber is a groupoid.

As I said in the beginning of this section, I think this condition is equivalent to saying that every morphism in the Étale space is cartesian (this should be an easy exercise).

Lemma 12.7. $p : \mathcal{E} \rightarrow \mathcal{B}$ is a fibration in groupoids if and only if every morphism in $\mathcal{E}$ is p -cartesian and any lifting problem

$$\begin{array}{ccc} & e & \\ & \downarrow & \\ b & \longrightarrow & p(e) \end{array}$$

has a solution.

Proof. The backward implication is easy. From the hypotheses we get immediately that p is a fibration. To show that the fibers are groupoids we lift an arbitrary morphism f and do lifting problem against itself. $\square$

Definition 12.8.

Lemma 12.9. Let X, Y be nerves of 1-categories. Then $p : X \rightarrow Y$ is a right fibration if and only if it is a fibration in groupoids.

Proof. The condition for right fibration for $n = 0$ gives existence, for $n = 1$ gives commutativity, for $n = 2$ gives uniqueness. $\square$

Now I put two other results:

Lemma 12.10. Kan fibration if and only if both left and right fibration.

Lemma 12.11. X is ∞ -groupoid if and only if $X \rightarrow \text{pt}$ is Kan fibration, if and only if it is left (or right) fibration.